



Automated AI-Powered Daily Email Summarization System

A report on Building Intelligent Agents using Agentic AI Platforms using Activepieces

As partial fulfillment of the requirement for the Course
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CERTIFICATE

Certified that the project entitled "Automated AI-Powered Daily Email Summarization System" carried out by **Shreenath Shenoy (1MS23CI112)**, **Preetam Lagishetty (1MS23CI093)**, **Saikiran Ponde (1MS23CI106)**, **Niranjan Kumar K (1MS23CI076)**, students of CSE(AI&ML) Department, **Ramaiah Institute of Technology, Bengaluru-54** partial fulfillment of the requirement for the Course Deep Learning & Computer Vision (23CI62) during the year 2026. The project work has been approved as it meets the academic requirements for the course.

Signature of the Guide

Signature of the HOD

ABSTRACT

This project presents an automated AI-powered daily email summarization system built using Activepieces. The system uses scheduled workflows to fetch recent emails from a user's inbox, filter and categorize them, and call an AI model to generate a concise, structured summary. Each digest includes prioritized communications, categories, and extracted actionable items so users can quickly understand the most important messages without reading every email. By integrating automation with intelligent analysis, the project reduces information overload, improves productivity, and demonstrates a practical application of agentic AI workflows for real-world communication management.

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Chapter 1

About Agentic AI Tool

1.1 n8n Tool

n8n is a powerful, low-code, node-based workflow automation tool that connects over 400 different apps and services to automate tasks. It allows users to build complex, multi-step workflows with visual triggers and actions, offering both a self-hosted option for data privacy and a managed cloud. n8n helps to connect any app with an API with any other and manipulate its data with little or no code.

- Customizable: highly flexible workflows and the option to build custom nodes.
- Convenient: use the npm or Docker to try out n8n, or the Cloud hosting option if you want us to handle the infrastructure.
- Privacy-focused: self-host n8n for privacy and security.

1.1.1 AI concepts in n8n

An AI agent builds on Large Language Models (LLMs). LLMs generate text based on input by predicting the next word. They can be used to select the best tool to achieve a task, or even simulate complex decision-making, but they can't act on decisions or use tools themselves. AI agents add goal-oriented functionality. They can use tools, act on their outputs, complete tasks and solve problems.

By incorporating the AI agent as a node, n8n can combine AI-driven steps with traditional programming for efficient, real-world workflows. For instance, simpler tasks, like

validating an email address, do not require AI, whereas a complex tasks, like processing the content of an email or dealing with multimodal inputs like images, audio), are excellent uses of an AI agent.

1.1.2 Built for AI

n8n has pivoted to become an AI-native platform, allowing you to build autonomous agents rather than just linear sequences.

- **Agentic Workflows:** Create agents that "think" and decide which tools to use (like searching a database vs. sending an email) based on a goal you set.
- **Built-in LangChain:** It has native nodes for LangChain, enabling you to chain together LLMs, memory, and custom tools visually.

1.1.3 Node Configuration

AI agents require a chat model to be attached to process the incoming prompts. Add a chat model by clicking the plus Add node icon button underneath the Chat Model connection on the AI Agent node -first connection along the bottom of the node. The search dialog will appear, filtered on 'Language Models'. These are the models with built-in support in n8n. For this tutorial we will use OpenAI Chat Model. Selecting the OpenAI Chat model from the list will attach it to the AI Agent node and open the node editor. One of the parameters which can be changed is the 'Model'. Note that for the basic OpenAI accounts, only the 'gpt-4o-mini' model is allowed.

1.2 Activepieces Tool

Activepieces is an open-source, no-code business automation agentic AI tool designed to allow users to connect various apps Google Sheets, Slack, and OpenAI apps into automated workflows called "flows".

1.2.1 Key Features

- **Open-Source & Self-Hostable:** Unlike proprietary competitors, Activepieces is built on an open ecosystem. You can self-host it for free on your own servers to maintain

total control over your data and avoid task limits.

- **AI-First Architecture:** It features native AI integrations and an "AI Copilot" that helps you design flows faster by suggesting steps based on natural language prompts.
- **Extensibility:** Developers can build custom integrations (called "pieces") using a TypeScript framework, which are then published as npm packages.
- **User Interface:** It uses a visual, drag-and-drop builder that is accessible to non-technical users while providing technical teams with advanced logic blocks like loops, branches, and custom code steps.

1.2.2 Technical Architecture & Self-Hosting

- **Core Components:** It consists of an App (API and UI), Workers that execute flows, an Engine that runs within a secure Sandbox, and Redis/PostgreSQL for task queuing and data storage.
- **Self-Hosting Requirements:** Hosted using Docker to ensure total data privacy and avoid task-execution costs.
- **Privacy:** On-premise deployment is ideal for regulated industries like healthcare or finance, as data never leaves your infrastructure.

1.2.3 AI-Powered Automation

Activepieces integrates AI directly into the building process and workflows.

- **AI Copilot:** An assistant can suggest steps or write custom TypeScript code based on natural language prompts.
- **Native AI Blocks:** It includes pre-built steps for OpenAI, Google Gemini, and Anthropic. These steps can summarize text, extract data from documents, or make complex decisions within a flow.
- **AI Agents:** Can build autonomous agents that follow a "Perceive, Think, Act" loop to handle end-to-end tasks like resolving customer tickets.

1.2.4 Advanced Logic & Control

- Human-in-the-Loop: You can insert approval steps that pause a flow until a human approves it via email or a dashboard.
- Custom Pieces: If an app isn't supported, developers can build a "piece" using the TypeScript SDK and publish it as an npm package.
- Complex Branching: Flows support routers (branching logic based on conditions), loops for processing lists, and error-handling steps.

1.3 Project Problem Statement

Email inboxes generate many messages every day, making it difficult for users to identify high-priority items and actionable tasks. Manual triage wastes time, increases the risk of missed communications, and contributes to information overload.

1.4 Proposed Project

This project builds an AI-powered daily email summarization system using Activepieces. It leverages scheduled workflows to retrieve recent emails, filter and preprocess content, and use an AI model to generate concise summaries. The system categorizes messages, assigns priority levels, and extracts actionable items, delivering a structured daily digest that helps users stay focused on what matters most.

1.5 Objectives

- Automate daily email retrieval and preprocessing using Activepieces.
- Generate concise daily summaries from recent inbox content.
- Categorize emails, assign priorities, and extract actionable items.
- Reduce user effort and improve productivity by surfacing critical communications.

Chapter 2

Introduction

2.1 AI Agents

AI agents are autonomous computational entities designed to perceive their environment, reason about it, and take appropriate actions to achieve specific goals. In the context of Agentic AI, these agents go beyond simple rule-based systems by incorporating advanced capabilities such as planning, decision-making, and learning. An AI agent typically consists of components like a perception module for understanding inputs (such as text or sensor data), a reasoning engine for analyzing situations, a memory system for retaining context and past experiences, and an action module to execute tasks. Modern AI agents often leverage techniques from machine learning, natural language processing, and reinforcement learning to operate effectively in dynamic environments. They can function independently or collaborate with other agents in multi-agent systems to solve complex problems. Applications of AI agents span across domains such as virtual assistants, healthcare, finance, robotics, and intelligent automation, making them a key building block in the development of next-generation intelligent systems.

2.2 Agentic AI

Agentic AI refers to a class of advanced artificial intelligence systems designed to operate through autonomous agents that can independently plan, reason, and take actions to achieve defined objectives. Unlike traditional AI models that primarily respond to direct inputs, Agentic AI systems exhibit goal-oriented behavior, where they can decompose com-

plex tasks into smaller steps, make decisions, and adapt dynamically based on feedback from their environment. These systems integrate key components such as intelligent agents, memory modules for context retention, planning and reasoning mechanisms, and tool-use capabilities that allow interaction with external systems like databases, APIs, or software applications. Agentic AI often leverages techniques from machine learning, natural language processing, and reinforcement learning to enhance its autonomy and effectiveness. It can function as a single-agent system or as a coordinated multi-agent framework, where multiple agents collaborate to solve complex, real-world problems. Due to its ability to automate decision-making and execute tasks with minimal human intervention, Agentic AI is increasingly applied in areas such as intelligent automation, personalized assistants, healthcare systems, and complex workflow management, representing a significant step toward more adaptive and self-directed AI systems.

2.3 Problem Statement

Email inboxes receive a large number of messages every day, making it difficult for users to quickly identify important emails, urgent requests, and actionable tasks. Manually reading and organizing emails consumes time, increases the chances of missing critical information, and contributes to information overload. Therefore, there is a need for an automated system that can analyze daily emails, filter relevant content, summarize key information, and highlight priorities and action items.

2.4 Proposed Project

The proposed project is an automated AI-powered daily email summarization system using Activepieces. The system uses a scheduled workflow to fetch recent emails from a user's inbox, filter and preprocess the messages, and send the relevant content to an AI model for summarization. The AI model generates a structured daily digest containing concise summaries, email categories, priority levels, and actionable items. This helps users save time, reduce manual effort, and focus on the most important communications.

Chapter 3

Project Workflow

3.1 Project architecture

The system is implemented as an Activepieces workflow that runs on a daily schedule. The flow retrieves recent emails from the user's inbox, applies filtering and preprocessing, classifies messages into categories, and extracts priorities and actionable items. The final step sends the consolidated content to an AI model to generate a structured summary digest.

The key components include:

- **Scheduled Trigger:** Starts the daily workflow at a configured time.
- **Email Retrieval:** Fetches recent messages from the inbox using an email connector.
- **Filtering and Preprocessing:** Removes spam, duplicates, and low-priority messages before summarization.
- **AI Summarization:** Sends prepared text to a language model with a prompt designed to extract summary, categories, priorities, and action items.
- **Output Delivery:** Produces a structured daily digest that can be displayed, stored, or sent to the user.

3.2 Software Requirements

- Operating System: Windows, Linux, or macOS
- Automation Platform: Activepieces
- Email Service: Gmail, Outlook, or any supported email provider
- Internet connection: Required for email access, Activepieces, and AI API calls
- Browser: Google Chrome, Microsoft Edge, or Firefox
- Optional Deployment Tools: Docker and Docker Compose for self-hosting Activepieces
- Database/Queue for Self-hosting: PostgreSQL and Redis
- Document Preparation: LaTeX / Overleaf / TeX Live for report generation

3.3 Hardware requirements

- Processor: Intel i3 / Ryzen 3 or higher
- RAM: Minimum 4 GB, recommended 8 GB
- Storage: Minimum 2 GB free space
- Internet connection: Required for email access, Activepieces, and AI API calls
- System: Laptop or desktop computer

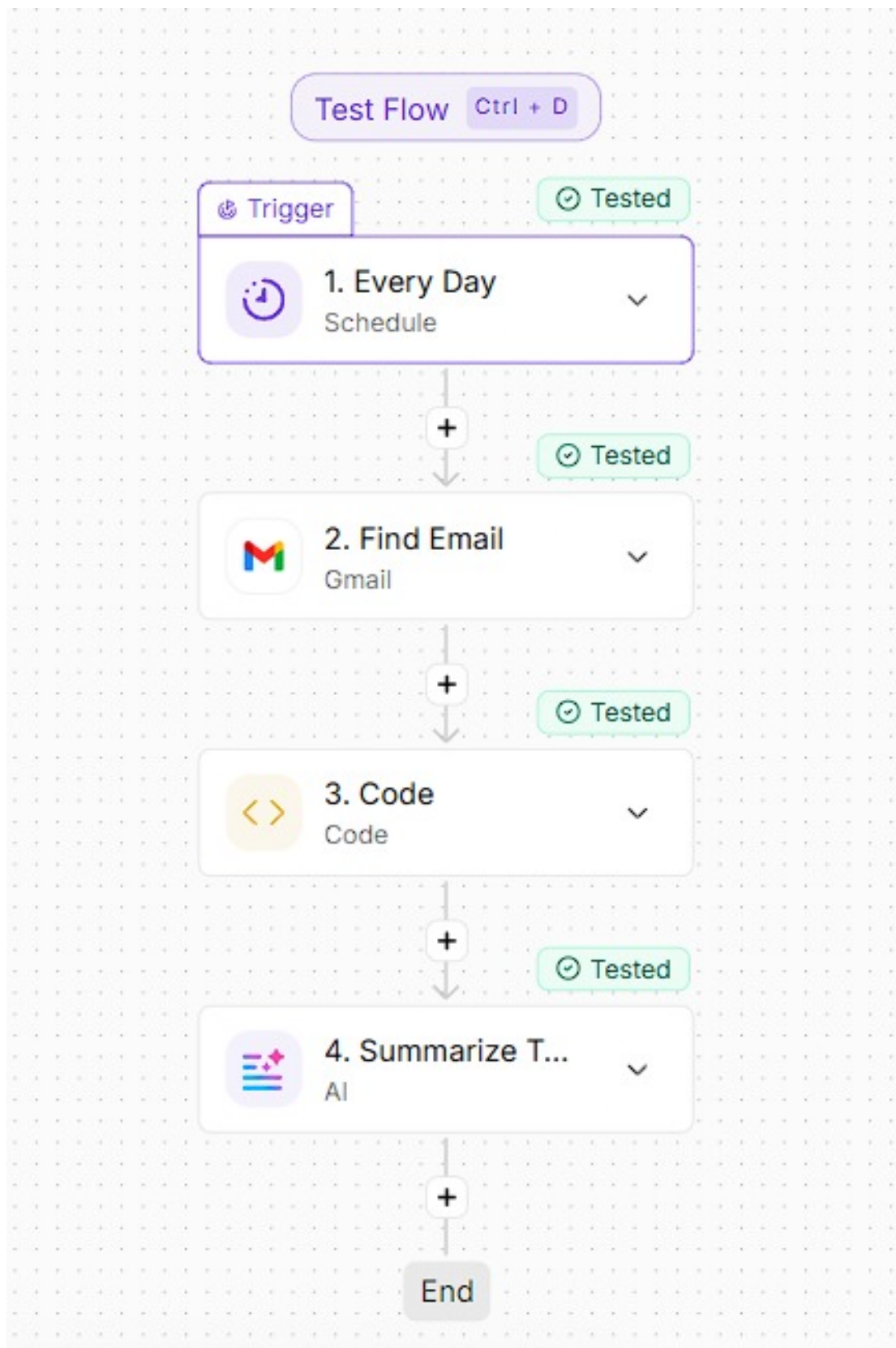


Figure 3.1: Activepieces workflow for automated email summarization

Chapter 4

Implementation

4.1 Activepieces Workflow Implementation

The email summarization system is implemented using Activepieces, a visual automation platform. The workflow begins with a scheduled trigger that runs daily, followed by steps that connect to the user's email inbox, retrieve recent messages, and apply initial filtering. Filtering removes irrelevant or duplicate emails and prepares content for the AI summarization step.

4.2 AI Summarization and Prompting

After preprocessing, the workflow consolidates the relevant email data and sends it to an AI model. The prompt is designed to ask the model to:

- Summarize the main content of each email.
- Categorize the email into defined communication groups.
- Assign a priority level based on urgency and importance.
- Extract actionable items that require follow-up.

This approach ensures the AI output is structured and useful for a daily digest.

4.3 Data Flow and Output

The final output includes a concise summary, categorized message groups, priority tags, and a list of actionable items. This structured summary can be delivered by email, stored in a document, or displayed in a dashboard for the user.

4.4 Deployment Considerations

The implementation is designed to run reliably in a self-hosted Activepieces environment for data privacy. It can be scheduled to execute automatically and extended with additional triggers or connectors as needed.

Chapter 5

Experiments and Results

5.1 Experimental Setup

The project was tested using an Activepieces flow connected to a sample email inbox. The workflow fetched recent email messages, applied filtering rules, and prepared the content for AI summarization. The summarization step used a prompt that asked the model to categorize emails, prioritize them, and identify actionable items.

5.2 Results

Figure 1 shows the raw email content before filtering and preprocessing. Figure 2 shows the state after filtering and prioritization. Figure 3 shows the final structured summary produced by the AI model, and Figure 4 shows the prompt design used for generation.

5.3 Discussion

The results demonstrate that the Activepieces workflow can efficiently prepare email content for AI summarization and that the generated digest helps users identify important communications, priorities, and action items quickly.

```

{ 2 items
  found: true
  results: { 2 items
    messages: { 16 items
      0: { 12 items
        id: "19d94d86f16007c9"
        attachments: [] 0 items
        headers: {} 0 items
        headerLines: 26 items
          [ 0 - 20 ]
          [ 20 - 26 ]
        html:
          "<div dir='auto'><div dir='auto'><br></div><div dir='auto'>Just finished my first day at the new office, and honestly... it was chaotic.</div>
          <div dir='auto'><br></div><div dir='auto'>I got lost twice trying to find the meeting room. And somehow, I ended up attending the wrong
          meeting for 20 minutes before realizing none of it made sense.</div><div dir='auto'><br></div><div dir='auto'>But the team seems nice. My
          manager, Kavya, is super sharp – a bit intimidating, though.</div><div dir='auto'><br></div><div dir='auto'>Let's see how this goes.</div>
          <div dir='auto'><br></div><div dir='auto'>- Arjun</div></div> "
        text:
          "Just finished my first day at the new office, and honestly... it was chaotic. I got lost twice trying to find the meeting room. And somehow,
          I ended up attending the wrong meeting for 20 minutes before realizing none of it made sense. But the team seems nice. My manager, Kavya, is
          super sharp – a bit intimidating, though. Let's see how this goes. - Arjun "
        textAsHtml:
          "<p>Just finished my first day at the new office, and honestly&mdash; it was chaotic.</p><p>I got lost twice trying to find the meeting room.
          And somehow, I ended up<br>attending the wrong meeting for 20 minutes before realizing none of it made<br>sense.</p><p>But the team seems
          nice. My manager, Kavya, is super sharp &mdash; a bit<br>intimidating, though.</p><p>Let&rsquo;s see how this goes.</p><p>&mdash;
          Arjun</p>"
        subject: "Hi Rohan,"
        date: "2026-04-16T05:51:31.000Z"
        to: { 3 items
          value: { 1 item
            0: { 2 items
              address: "1ms23c112@msrit.edu"
              name: ""
            }
          }
        }
      }
    }
  }
}

```

Figure 5.1: Email content before filtering and preprocessing

```

Output
{ 1 item
  formatted_text:
    "📧 Emails (5) 1. Hi Rohan, - Just finished my first day at the new office, and honestly... 2. Security alert - A new sign-in on Windows
    1ms23c112@msrit.edu We noticed 3. 16 Coursera Courses for Today's Fastest-Moving Tech Trends - Dear Shreenath Shenoy Shenoy, The tech
    landscape is moving 4. Competition Launch: The IJCAI-ECAI 2026 NeuroGolf Challenge - Kaggle Hi Shreenath Shenoy MSRIT, What does it real 5.
    [ACTION REQUIRED] New issue in Summarize YouTube Videos from Transcript to Google Docs - "
}

```

Figure 5.2: Email content after filtering and prioritization

```

Output
📧 Daily Email Report

1. From: Google <no-reply@accounts.google.com>
   Subject: Security alert
   Summary: New sign-in detected on a Windows device for your Google account.
   Priority: HIGH
   Category: Personal
   Action: Yes – Review account activity and secure the account if the sign-in wasn't you.

2. From: Activepieces <noreply@activepieces.com>
   Subject: [ACTION REQUIRED] New issue in Summarize YouTube Videos from Transcript to Google Docs
   Summary: An automation flow has an issue and needs to be fixed in the dashboard.
   Priority: HIGH
   Category: Work
   Action: Yes – Open Runs → Issues, fix the flow, and mark it resolved.

```

Figure 5.3: Final AI-generated email summary digest

Prompt *

You are an intelligent email assistant.
Analyze the following emails and create a structured report.
For each email:

- Sender
- Subject
- Summary (1 line)
- Priority (HIGH / MEDIUM / LOW)
- Category (Work / College / Personal / Other)
- Action Required (Yes/No + what action)

Format:

 Daily Email Report

1. From: 😊
Subject: 😊
Summary: 😊
Priority: 😊
Category: 😊
Action: 😊

At the end include:

 High Priority Emails:
(list)

 Action Items:
(list clearly)

Ignore spammy or promotional content.
Emails:

Figure 5.4: AI prompt used for structured summary generation

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project demonstrates how Activepieces can automate email retrieval and combine it with AI summarization to reduce information overload and improve productivity. By categorizing messages, assigning priorities, and extracting actionable items, the system helps users focus on important communications without reading every message.

6.2 Future Scope

The project can be extended to support multiple inbox providers, add natural language query access to summaries, integrate calendar and task management systems, and include multilingual email summarization and attachment handling.

Bibliography

- [1] A. Singh, A. Ehtesham, S. Kumar and T. T. Khoei, "Enhancing AI Systems with Agentic Workflows Patterns in Large Language Model," 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 527-532, doi: 10.1109/AI-IoT61789.2024.10578990.
- [2] D. B. Acharya, K. Kuppan and B. Divya, "Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey," in IEEE Access, vol. 13, pp. 18912-18936, 2025, doi: 10.1109/ACCESS.2025.3532853.
- [3] S. Chahar, P. Gaba, K. S. Kaswan, J. S. Dhatteval and A. Baliyan, "Agentic AI: A Paradigm Shift in Autonomous Decision-Making and Intelligent Systems," 2025 Artificial Intelligence and Smart Technologies for Sustainability Conference (AISTS), Rajkot, India, 2025, pp. 1-6, doi: 10.1109/AISTS66100.2025.11232814.
- [4] C. Shah and R. W. White, "Agents Are Not Enough," in Computer, vol. 58, no. 10, pp. 87-92, Oct. 2025, doi: 10.1109/MC.2025.3575829.
- [5] Huang, Yu. "Levels of ai agents: from rules to large language models." arXiv preprint arXiv:2405.06643 (2024).
- [6] <https://www.activepieces.com/>
- [7] <https://n8n.io/>